Synthetic Homotopy Theory

Abbreviated lecture notes

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Lecture 1

What is homotopy theory?

Homotopy theory is the study of paths between.

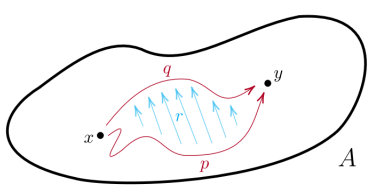
What is synthetic homotopy theory?

Synthetic homotopy theory is the study of the identity type in Martin-Löf Type Theory.

What is the connection?

Awodey, Hoffmann, Streicher, Voevodsky, Warren[1]:

- Turns out: types in type theory can be interpreted as spaces
  - 'A is a type' (or $A : \mathcal{U}$) —> 'A is a topological space (up to *homotopy*, i.e. continuous deformation, but we don't need to worry about this now)'
  - $x : A$ —> 'x is a point in A.
  - $p : x =_A y$ —> 'p is a (continuous) path from x to y in A'.
  - $r : p =_{x=y} q$ —> 'p is a *higher* path/continuous deformation/*homotopy* from p to q'



While all paths may be interesting to homotopy theorists, there are certainly quite a lot of them... One thing that homotopy theorists like to do is to restrict to a certain classs of paths, namely those that are *loops*, i.e. start and end on the same point.

[1] Voevodsky is the person who is directly attributed this connection, but the other authors have also made crucial contributions that inspired this interpretation

Draw pic

Given a space A and $x \in A$, we can define

$$\pi_1(A, x) := \{\text{Loops in } A \text{ starting and ending at x}\}/\sim$$

where we quotient out by *homotopy* – two loops are equal if one can be continuously deformed into the other. The reason that we restrict to loops is that this gives us a group structure. Indeed, given two loops p and q, we can compose these ('walk along p and then along q'). Similarly, we can revert them: p^{-1} = *'p, but backwards'*. We also get a neutral element const_x which is the constant loop standing still on x.

The group $\pi_1(A, x)$ is called the *fundamental group* of A (based at x). Now, why stop at 1-dimensional loops from x to itself? We can easily (although we won't do it here) define the notion of an n-dimensional loop and can similarly define

$$\pi_n(A, x) = \{n\text{-dimensional loops in } A \text{ touching } x\}/\sim$$

where we again quotient by an appropriate notion of homotopy. This is called the *nth homotopy group of A (based at x)*. Knowing $\pi_n(A, x)$ a space A and suitable $x \in A$ for all n esentially tells you 'all you need to know' about your space (if you're a homotopy theorist, that is). Unfortunately, these are *incredibly hard* to compute in general. For this reason, homotopy theorists often focus on the special case when A is the m-dimensional (hyper-)sphere S^m. In a way, all spaces can ('up to homotopy') be thought of as consisting of a bunch of spheres of varying dimension glued together, this seems like a very suitable thing to study. Unfortunately, also computing $\pi_n(S^m, \star)$ for arbitrary $n, m \geq 1$ is notoriously hard – or perhaps this is fortunate, since computing these has, in a way, become the raison d'etre of homotopy theory.

The goal of this course will be to explore whether we can compute any of these $\pi_n(S^m, \star)$ in synthetic homotopy theory, i.e. using the language of dependent type theory or, actually, *Homotopy Type Theory* (HoTT), an extension of this language). In thislecture, we will define homotopy groups in HoTT. In lecture 2, we introduce some actual spaces into our type theory (in particular spheres). In lectures 3 and 4 we will compute some homotopy groups of these spaces – in particular of the circle and the 3-sphere.

Note, that HoTT is just dependent type theory with *univalence* and *higher inductive types*. I'll introduce these concepts as we go.

Paths

Let us recall what is probably the most important notion from dependent type theory:

Def (Path induction)**.** Let $x : A$. In order to prove a statement/construct an element of type $(y : A)(p : x = y) \to B(y, p)$, it suffices to prove/construct an element of $B(x, \mathrm{refl}_x)$.

Let's give some examples of this principle in action:

Ex (Transport)**.** Given $p : x =_A y$ and $B : A \to \mathcal{U}$, we get a map $\mathrm{transport}^B_p : B(x) \to B(y)$ by path induction on p:

$$\mathrm{transport}^B_{\mathrm{refl}_x} : B(x) \to B(x) \quad := \quad \mathsf{id}_{B(x)}$$

draw picture explaining transport

Ex (Function application)**.** Given $f : X \to Y$ and $p : x = y$, we have $\mathrm{ap}_f(p) : f(x) = f(y)$ (by path induction on p).

Question. Function application is only stated for plain functions, but we're in dependent type theory. What if $f : (x : X) \to B(x)$ is a dependent function?

We can't write $\mathrm{ap}_f(p) : f(x) = f(y)$ because $f(x) : B(x)$ and $f(y) : B(y)$, so this equality doesn't type check! But $B(x)$ and $B(y)$ are equal types (by p), so we should still be able to express the idea that $f(x)$ and $f(y)$ are 'equal' (modulo p):

Def (Dependent paths)**.** Given $B : X \to \mathcal{U}$ dep. type, $p : x_0 = x_1$, and terms $b_0 : B(x_0)$ and $b : B(x_1)$, define

$$(b_0 =^B_p b_1) := (\mathrm{transport}^B_p(b_0) = b_1)$$

Note that the above definition coincides *exactly* with the usual identity type whenever p is refl.

Ex (Function application)**.** Given $f : X \to Y$ and $p : x = y$, we have $\mathrm{apd}_f : (p : x = y) \to f(x) =^B_p f(y)$ (by path induction on p).

Remember that homotopy theorists were interested in composing paths and inverting paths? In HoTT, this idea simply corresponds to the fact that our identity type is transitive and symmetric.

Ex (Path composition/inversion)**.** Given $p : x = y$ and $q : y = z$, we can define $p \cdot q : x = z$ and $p^{-1} : y = z$ by path induction on p. These operations satisfy (for suitable p, q and r):

- $p \cdot p^{-1} = \mathrm{refl}$
- $p^{-1} \cdot p = \mathrm{refl}$
- $p \cdot \mathrm{refl} = p$
- $\mathrm{refl} \cdot p = p$
- $(p \cdot q) \cdot r = p \cdot (q \cdot r)$

Are we there yet?

So, we have path types with notions of composition, inversion, and a neutral element. Why not just play the same game as in traditional homotopy theory and set $\pi_1(X, x) := (x = x)$ with group operations given by path composition and path reversal. This is *almost* right. The issue: the group laws hold but they can hold in *infinitely many ways*! Suppose you and I both are tasked with defining $\pi_1(X, x)$ in type theory. Both of us provide the same underlying type, the same operations, and the same neutral element – but we have provided different proofs of the fact that, say, $p \cdot p^{-1} = \mathrm{refl}$. Which definition is the appropriate one? Mine or yours? To arrive at a well-defined notion of groups, we would like to make sure that this situation never arises – we would like to 'quotient' by all 'higher paths' (i.e. paths involving other paths, such as the path witnessing $p \cdot p^{-1} = \mathrm{refl}$. The next part of this lecture is about how to construct these kinds of quotients.

n-types

In order to even state that we wish to 'quotient out higher paths', we need a way to pinpoint what we even mean by 'higher paths'. A good way of capturing a type having 'higher structure' in general is the notion of an n-type. Let us start with two special cases:

Def. A type X is **contractible** if the type

$$\mathrm{isContr}(X) := (x : X) \times ((y : X) \to x =_X y)$$

is inhabited.

Def. A type X is a **proposition** if the type

$$\mathrm{isProp}(X) := (x, y : X) \to x =_X y$$

is inhabited.

Usin this, we can define what it means to be an n-type:

Def. A type X is

- a **(-2)-type** if $\mathrm{isContr}\, A$,
- a **(-1)-type** if $\mathrm{isProp}\, A$, and
- an **n-type** if $(x, y : X) \to \mathsf{is}\text{-}(n-1)\text{-}\mathsf{type}\ (x =_A y)$.

We may use the following conventions

Convention.

- 0-types are called **Sets**

- 1-types are called **Groupoids**

To get some intuition (hopefully, you will get even more later on in the course), let us give some 'concrete' examples.

Ex.

- All (-2)-types:
  - The unit type 1. Actually, all (-2)-types are equivalent to this type.
  - The **singleton** type (over some $x : X$) i.e. $(y : X) \times (x = y)$ is contractible.
- Propositions ((-1)-types):
  - $1, \emptyset$ (and everything in between)
  - The statement that a type is an n-type is itself a (-1)-type: $\mathsf{isProp}(\mathsf{is}\text{-}n\text{-}\mathsf{type}\ (X))$.
  - The **singleton** type (over some $x : X$) i.e. $(y : X) \times (x = y)$ is contractible.
  - We haven't come this far yet, but we will soon define what it means for a function to the an equivalence (roughly, 'invertible'). The statement that $f : A \to B$ is an equivalence is a (-1)-type.
- Sets (0-types). These are types that are discrete in nature – they can have a bunch of points but no interesting paths (i.e. no interesting homotopical structure):
  - Any (plain) inductive type, e.g. $\mathbb{Z}$
  - cf. Hedberg's theorem if you're interested in how to prove this
- n-type:
  - The universe of n-types, $\mathcal{U}^{\leq n} := (X : \mathcal{U}) \times \mathsf{is}\text{-}n\text{-}\mathsf{type}\ (X)$, is an $(n+1)$-type.
  - Any n-type is also an $(n+1)$-type.

We said before that we'd like to be able to 'quotient' by all 'higher structure' of a type. In other words, we would like to be able to take a type and turn it into an n-type. This is precisely the role of the n-truncation. Here, it is stated as a definition, although we never show how it's defined... For now, take it for granted:

Def. Given a type X, there is an n-type $\|X\|_n$ equipped with a map $|-|_n : X \to \|X\|_n$ s.t. for every $f : X \to Y$ where Y is an n-type, we have

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "|-|_n"'] & Y \\ \|X\|_n \arrow[ur, dashed] & \end{tikzcd}$$

Alternatively:

$$_ \circ |_|_n : (\|X\|_n \to Y) \to (X \to Y)$$

is an equivalence. This phrasing might be better because we easily are able to generalise it to get the induction principle (i.e. dependent elimination) for any dependent (n-)type $B : X \to \mathcal{U}^{\leq n}$: we require that

$$_ \circ |_|_n : ((x : \|X\|_n) \to B(x)) \to ((x : X) \to B(|x|_n))$$

is an equivalence.

- Common notation: $\|X\| := \|X\|_{-1}$

Finally: Homotopy groups

Def. A **pointed type** is a pair $(X, \star_X) : \underbrace{(X : \mathcal{U}) \times X}_{\mathcal{U}_\star}$ where $X : \mathcal{U}$ and $\star_X : X$.

Def. A **pointed map** is a pair $(f; \star_X) : \underbrace{(f : X \to Y)(\times f(\star_X) = \star_Y)}_{(X,\star_X) \to_\star (Y,\star_Y)}$

Notation.

- Often simply write $X : \mathcal{U}_\star$ and $f : X \to_\star Y$, i.e. we leave basepoints and pointedness proofs implicit
- Always write $\star_X : X$ for basepoints and $\star_f : f(\star_X) = \star_Y$ for pointedness proofs.

Def. We define the **loop space** of a pointed types X by

$$\Omega(X) := (\star_X = \star_X)$$

This type is itself pointed by $\mathrm{refl}_{\star_X}$.

- It is **functorial**: given $f : X \to_\star Y$, we define $\Omega(f) : \Omega(X) \to \Omega(Y)$ by

$$\Omega(f)(p) := (\star_f)^{-1} \cdot \mathrm{ap}_f(p) \cdot \star_f$$

- Path composition defines a 'group structure' on Ω
  - Problem: group laws only hold up to homotopy.
    - For instance there might be more than one element of $p \cdot p^{-1} = \mathrm{refl}_{\star_X}$.
  - Solution: need to truncate

Def. The **first homotopy group** (also the **the fundamental group**) of a pointed type X is defined by $\pi_1(X) := \|X\|_0$

- Multiplication is inherited immediately from path composition on $\Omega(X)$. By the univeral property: $|p|_0 \cdot |q|_0 = |p \cdot q|_0$

Note $\Omega : \mathcal{U}_\star \to \mathcal{U}_\star$ – can be iterated.

Def. The nth homotopy group of $X : \mathcal{U}_\star$ is defined by

$$\pi_n X := \|\Omega^n X\|_0$$

Note that $n \geq 1$!

- Ω is functorial $\implies$ π_n is functorial
- Write $f^* := \pi_n(f) : \pi_n(X) \to \pi_n(Y)$ for its functorial action.

Things I should've talked about but didn't

I didn't have time to mention that there is a notion of n-connected types. This notion is, in a sense, dual to the notion of n-type:

Def. A type X is **n-connected** if $\|X\|_n$ is contractible.

We often simply write 'connected' instead of '0-connected'.

Proposition. *If X is $(n \geq -1)$-connected, then $x =_X y$ is $(n-1)$-connected for all $x, y : X$.*

Just like types, functions can also be 'n-types' (we call these n-truncated) and 'n-connected'. This essentially means that they behave nicely when mapping into/out out of n-types/n-connected types. To define this notion, we need fibres:

Def. The **fibre** of a function $f : X \to Y$ over a point $y : Y$ is the type

$$\mathrm{fib}_f(y) := (x : X) \times (f(x) = y)$$

Def. A map $f : X \to Y$ is:

1. **n-truncated** if $\mathrm{fib}_f(y)$ is an n-type for all $y : Y$,
2. **n-connected** if $\mathrm{fib}_f(y)$ is n-connected for all $y : Y$.

These notions are crucial, but their significance is perhaps most easily illustrated via the following results (our very first homotopy group computations!):

Proposition. *If X is n-type, then $\pi_k(X)$ vanishes $\forall k > n$.*

Corollary. *$\pi_{n\geq 1}(X)$ vanishes for any set X. In particular for $X = 1$.*

Proposition. *If X is n-connected, then $\pi_k(X)$ vanishes $\forall k \leq n$.*

Recap: univalence

I never stated the univalence axiom, and that would probably have been appropriate for a course basde on homotopy type theory. To state it, we need to define equivalences:

Def. A function $f : X \to Y$ is an **equivalence** if $\mathrm{fib}_f(y)$ is contractible for all y.

- Let $(X \simeq Y) \times \mathrm{isEquiv}(f)$
- Fact: $\mathrm{id}_X : X \to X$ is an equivalence.

Remark. Why not use $\underbrace{(g : (Y \to X)) \times (g \circ f = \mathrm{id}) \times (f \circ g = \mathrm{id})}_{:=\mathrm{qinv}(f)}$ as definition?

The issue with this definition is that it is *not a proposition*, so there may be many different proofs of it for a given function f. This makes it inappropriate to use. However, it turns out that $\mathrm{qinv}(f)$ and $\mathrm{isEquiv}(f)$ both imply each other (i.e. they are *logically equivalent*), so we can just pretend that these definitions are the same. That's certainly what I do when I construct (most) equivalences. For this reason, I will often talk about the 'inverse' of an equivalence.

Axiom (Univalence)**.** *The canonical map $X =_{\mathcal{U}} Y \to (X \simeq Y)$ is an equivalence. In particular, $X \simeq Y \to X = Y$*

Above, the 'canonical map' is defined using an appropriate transport or by path induction. Constructing it is a good exercise if you feel uncertain about these concepts.

Fact. Univalence implies function extensionality, i.e. the statement that if $(((xy : X) \to f(x) = g(x))$ then $f = g$

From now on, we will always assume (and use) univalence/functio nextensionality, often without comment.

Lecture 2: A menagerie of higher inductive types

Plan

1. Introduce basic HITS: The circle, pushouts, suspensions and n-spheres
2. Compute vanishing homotopy groups of spheres
3. 3x3-lemma

The circle

- Higher inductive types (HITs): inductive types with path constructors

Def. The **circle**, S^1,is the HIT generated by

$$\begin{aligned} \star_{S^1} &: S^1 \\ \mathrm{loop} &: \star_{S^1} = \star_{S^1} \end{aligned}$$

Draw it

Elimination principle: To define a map $f : S^1 \to X$, need to define:

- $f(\star_{S^1}) :=? \quad : \quad X$
- $\mathrm{ap}_f(\mathrm{loop}) :=? \quad : \quad f(\star_{S^1}) = f(\star_{S^1})$

Alt: the evaluation map

$$(S^1 \to X) \to (x : X) \times (x =_X x), \quad f \mapsto (f(*), \mathrm{ap}_f(\mathrm{loop}))$$

is an equivalence.
Dependent version: for a dependent type $B : S^1 \to \mathcal{U}$, the evaluation map

$$\Big((z : S^1) \to B(z)\Big) \to \Big(x : B(*) \times (x =^B_{\mathrm{loop}} x)\Big), \quad f \mapsto (f(*), \mathrm{apd}_f(\mathrm{loop}))$$

is an equivalence.

Pushouts

Can define almost any space CW complex we want this way, but almost all of these constructions will turn out to be equivalent to an instance of the following:

Def. Given a span $B \xleftarrow{f} A \xrightarrow{g} C$ we can form its **pushout** by the higher inductive type $B \sqcup^A C$ with the following constructors:

$$\begin{aligned} \mathrm{inl} &: B \to B \sqcup^A C \\ \mathrm{inr} &: C \to B \sqcup^A C \\ \mathrm{glue} &: (a : A) \to \mathrm{inl}(fa) = \mathrm{inr}(ga) \end{aligned}$$

Draw pictures

We write

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d] \\ B \arrow[r] \arrow[ur, phantom, "\ulcorner"] & D \end{tikzcd}$$

to express that D is the pushout $B \sqcup^A C$ defined above.

Def. Numerous construction of (in general, higher) types as pushouts:

1. ΣX, the **suspension** of X; = pushout of $1 \leftarrow X \rightarrow 1$. Conventions:
   - $N := \mathrm{inl}(\star_1)$
   - $S := \mathrm{inr}(\star_1)$, and
   - $\mathrm{merid}(x) := \mathrm{glue}(x)$
   - Always take ΣX to be pointed by N.
2. $X * Y$, the **join**, of X and Y = pushout of $X \leftarrow X \times Y \rightarrow Y$.
3. $X \vee Y$, the **wedge sum** of two <u>pointed types</u> X and Y = pushout of $X \xleftarrow{\star_X} 1 \xrightarrow{\star_Y} Y$.
4. Given a map $f : X \to Y$ its **cofibre**, $\mathrm{cofib}(f)$, is given by the pushout of the diagram $1 \leftarrow X \rightarrow Y$.
5. There is an inclusion $\iota : X \vee Y \to X \times Y$ defined by

$$\begin{aligned} X \vee Y &\to X \times Y \\ \mathrm{inl}\, x &\mapsto (x, y_0) \\ \mathrm{inr}\, y &\mapsto (x_0, y) \end{aligned}$$

   Its cofibre is called the **smash product**, denoted $X \wedge Y$.

Suspensions/spheres

We have seen We will use the above to define spheres

- Have seen S^1
- Q: How to define S^n?
- Easy for fixed n but in general?

Solution: iterated suspensions.

An immediate use for suspensions is given by defining higher dimensional spheres. Although it is clear how to define each sphere individually as an HIT, this doesn't give a definition of all the spheres internal to type theory, as we are unable to produce a function $\mathbb{N} \to \mathcal{U}$ this way. Instead we can define the higher spheres uniformly via iterated suspension.

Def. We define the n-sphere, S^n, by $S^n := \Sigma\emptyset$.

Draw

Theorem (Loop Space–Suspension Adjunction)**.** *Let X and Y be a pointed type. We have*

$$(\Sigma X \to_\star Y) \simeq (X \to_\star \Omega(Y))$$

Proof. Unfold universal property of ΣX:

$$\begin{aligned}
(\Sigma X \to_\star Y) &\equiv (f : \Sigma X \to Y) \times (f(N) = \star_Y) \\
&\simeq (y_N : Y) \times (y_S : Y) \times (X \to y_N = y_S) \times (y_N = \star_Y) \\
&\simeq \underbrace{(y_N : Y) \times (y_N = \star_Y)}_{\text{contr. at } (\star_Y, \text{refl})} \times (y_S : Y) \times (X \to y_N = y_S) \\
&\simeq (y_S : Y) \times (X \to \star_Y = y_S) \\
&\simeq (y_S : Y) \times (p : X \to \star_Y = y_S) \times (q : \star_Y = y_S) \times (q = p(\star_X)) \\
&\simeq \underbrace{(y_S : Y) \times (q : \star_Y = y_S)}_{\text{contr. at } (\star_Y, \text{refl})} \times (p : X \to \star_Y = y_S) \times (q = p(\star_X)) \\
&\simeq (p : X \to \star_Y = \star_Y) \times (\text{refl} = p(\star_X)) \\
&\simeq (X \to_\star \Omega Y)
\end{aligned}$$

□

- The unit $X \to_\star \Omega(\Sigma X)$ we will denote by σ. Very important map!
- **Exercise** Trace the above proof to verify that $\sigma(x) = \text{merid}(x) \cdot \text{merid}(x_0)^{-1}$

Recall following exercise:

Exercise A type A is n-connected iff the diagonal map $X \to (A \to X)$ is an equivalence for any n-types X.

Theorem (Connectivity of suspension)**.** *If X is n-connected then ΣX is $n+1$-connected.*

Proof. Using the exercise, it suffices to show that the diagonal map

$$Y \to (\Sigma X \to Y)$$

is an equivalence for all $n+1$-types Y. We give an equivalence $(\Sigma X \to Y) \simeq Y$, and leave it to the reader to check it is compatible with the diagonal map:

$$\begin{aligned}(\Sigma X \to Y) &\simeq (y_0, y_1 : Y) \times (X \to y_0 = y_1)\\ &\simeq (y_0, y_1 : Y) \times (y_0 = y_1) \simeq Y\end{aligned}$$

Here we used that $y_0 = y_1$ is an n-type and the converse of the cited exercise. □

(There's an alternative proof using the fact that n-connected maps are closed under pushouts.)

Corollary. *The sphere S^n is $n-1$-connected.*

The three-by-three lemma and applications

Given a commutative diagram

$$\begin{tikzcd} B & A \arrow[l] \arrow[r] \arrow[d, dashed, "h"] & C \arrow[d, "g"] \\ Y \arrow[from=u, "f"'] & X \arrow[l] \arrow[r] & Z \end{tikzcd}$$

we get an induced map $B \sqcup^A C \to Y \sqcup^X Z$ **exercise: construct it**.
Consider:

$$\begin{tikzcd} A_{00} & A_{02} \arrow[l] \arrow[r] & A_{04} \\ A_{20} \arrow[u] \arrow[d] & A_{22} \arrow[l] \arrow[r] \arrow[u] \arrow[d] & A_{24} \arrow[u] \arrow[d] \\ A_{40} & A_{42} \arrow[l] \arrow[r] & A_{44} \end{tikzcd}$$

including homotopies for the four squares, we can either form the pushouts of the columns resulting in a span

$$A_{\bullet 0} \longleftarrow A_{\bullet 2} \longrightarrow A_{\bullet 4},$$

or form the pushouts of the rows resulting in a span

$$A_{0\bullet} \longleftarrow A_{2\bullet} \longrightarrow A_{4\bullet}.$$

Lemma (3×3)**.** *There is a natural equivalence between the pushouts of the two spans above.*

Proof. Direct but technical. □

Let's consider an application. For it, we need a quick lemma whose proof is direct:

Lemma. *Given an equivance $e : A \simeq B$, the following is a pushout square:*

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "e"'] & C \arrow[d] \\ B \arrow[r] \arrow[ur, phantom, "\ulcorner"] & C \end{tikzcd}$$

Proposition. *Consider a commutative diagram of the form:*

$$\begin{tikzcd} A \arrow[r] \arrow[d] & C \arrow[r] \arrow[d] & E \arrow[d] \\ B \arrow[r] \arrow[ur, phantom, "\ulcorner"] & D \arrow[r] & F \end{tikzcd}$$

TFAE:

- *F is a pushout of the rightmost square*
- *F is a pushout of the outermost square*

Proof.

$$\begin{tikzcd} B & \emptyset \arrow[l] \arrow[r] & \emptyset \arrow[r, equal] & B \\ A \arrow[u] \arrow[d] & \emptyset \arrow[l] \arrow[u] \arrow[r] \arrow[d] & \emptyset \arrow[r, equal] \arrow[u] \arrow[d] & A \\ C \arrow[d, equal] & C \arrow[l] \arrow[r] \arrow[d, equal] & E \arrow[r, equal] \arrow[d, equal] & E \\ D & C & E & \end{tikzcd}$$

□

Proposition. *For any types A, B, C, there is a natural equivalence*

$$(A * B) * C \simeq A * (B * C).$$

To prove this, we will need to know that the pushouts commute with products, i.e. if

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d] \\ B \arrow[r] \arrow[ur, phantom, "\ulcorner"] & D \end{tikzcd}$$

is a pushout square, then so is

$$\begin{tikzcd} A \times X \arrow[r, "g \times \mathsf{id}_X"] \arrow[d, "f \times \mathsf{id}_X"'] & C \times X \arrow[d] \\ B \times X \arrow[r] \arrow[ur, phantom, "\ulcorner"] & D \times X \end{tikzcd}$$

(exercise).

Proof of proposition. Consider the commuting diagram

$$\begin{tikzcd} A & A \times B \arrow[l] \arrow[r] & B \\ A \times C \arrow[u] \arrow[d, "\sim"] & A \times B \times C \arrow[l] \arrow[r] \arrow[u] \arrow[d] & B \times C \arrow[u] \arrow[d] \\ A \times C & A \times C \arrow[l, "\sim"'] \arrow[r] & C \end{tikzcd}$$

The span given by taking the pushouts of the columns is

$$A \longleftarrow A \times (B * C) \longrightarrow B * C,$$

whose pushout is $A * (B * C)$. The span given by taking the pushouts of the rows is

$$A * B \longleftarrow (A * B) \times C \longrightarrow C,$$

whose pushout is $(A * B) * C$. □

In the exercises we'll construct natural equivalences $2 * A \simeq \Sigma A$.

Corollary. *For every $n, m : \mathbb{N}$ we have an equivalence*

$$S^n * S^m \simeq S^{n+m+1}.$$

Lecture 3: The Hopf fibration and the long exact sequence

The fundamental group of the circle

plan

- Compute our first proper homotopy group: $\pi_1(S^1)$.
- Main tool: Long exact sequence of homotopy groups
- Compute $\pi_3(S^2)$

—

- Spoiler: $\pi_1(S^1) \cong \mathbb{Z}$
- S^1 is special : already $\Omega S^1 \simeq \mathbb{Z}$ (no need to truncate)

Lemma. *Let $P : S^1 \to \mathcal{U}$ be a family of propositions. If $P(\star_S)$ holds then $P(x)$ holds for all $x : S^1$.*

Proof. S^1 is 0-connected, so $S^1 \to (S^1 \to B(x)))$ □

'digression': some transport lemmas

Lemma. *Let $P, Q : X \to \mathcal{U}$ and $p : x_0 =_X x_1$. Let $f : P(x_0) \to Q(x_0)$. Then we can identify* $\mathrm{transport}_p^{x \mapsto P(x) \to Q(x)}(f) : P(x_1) \to Q(x_1)$ *with the composition*

$$\mathrm{transport}_p^Q \circ f \circ \mathrm{transport}_{p^{-1}}^Q : P(x_1) \to Q(x_1).$$

Proof. By path induction. □

Lemma. *Let $p : x = y$ and $q : y = z$. We have* $\mathrm{transport}_q^{b \mapsto x = b}(p) : p \cdot q$.

Proof. By path induction. □

We can easily define a map $\mathrm{loop}^{(-)} : \mathbb{Z} \to \Omega S^1$ by $\mathrm{loop}^{n+1} := \mathrm{loop}^n \cdot \mathrm{loop}$.

Q: Other direction?? $\Omega S^1 \to \mathbb{Z}$? Get stuck

Change approach: fundamental theorem of identity types Suppose we could define a family $R : S^1 \to \mathcal{U}$ such that:

- $R(\star_{S^1}) = \mathbb{Z}$
- $(x : S^1) \times R(x)$ is contractible

We get $\underbrace{\star_{S^1} = \star_{S^1}}_{\Omega S^1} \to \underbrace{R(\star_{S^1})}_{\mathbb{Z}}$ is an equivalence.

Defining R We define R by S^1-elimination:

$$\begin{aligned} R : S^1 &\to \mathcal{U} \\ R(*) &:= \mathbb{Z} \\ \mathrm{ap}_R(\mathrm{loop}) &:= \mathsf{ua}(x \mapsto x + 1) \end{aligned}$$

draw picture to argue why contr.

Fact: $\mathrm{transport}^R_{\mathrm{loop}}(x) = x + 1$ and $\mathrm{transport}^R_{-\mathrm{loop}}(x) = x - 1$.

Proving contractibility **Point of contraction:** there is a canonical point of $(x : S^1) \times R(x)$, namely $(\star_{S^1}, 0)$
Uniqueness: Given any other two points $x : S^1$ and $r : R(x)$, show $(\star_{S^1}, 0) = (x, y)$.

- i.e. provide two pieces of data: $p : (x : S^1)(r : R(x)) \to \star_{S^1} = x$ and $q : (x : S^1)(r : R(x)) \to 0 =^R_{p(x,r)} r$. We leave q as an exercise.

We construct p by S^1-elimination:

$$\begin{aligned} p(\star_{S^1}) &: (r : \mathbb{Z}) \to \star_{S^1} = \star_{S^1} \\ p(\star_{S^1})(r) &= \mathrm{loop}^r \end{aligned}$$

For the loop case, need to show

$$(r \mapsto \mathrm{loop}^r) =^{x \mapsto R(x) \to \star_{S^1} = x}_{\mathrm{loop}} (r \mapsto \mathrm{loop}^r)$$

i.e. for all $r : \mathbb{Z}$

$$\mathrm{transport}^{x \mapsto R(x) \to \star_{S^1} = x}_{\mathrm{loop}}(r \mapsto \mathrm{loop}^r)(r) = \mathrm{loop}^r$$

Decompose transport on LHS:

$$\mathrm{transport}^{x \mapsto \star_{S^1} = x}_{\mathrm{loop}}(\mathrm{loop}^{\mathrm{transport}^R_{\mathrm{loop}^{-1}}(r)})$$

We compute

$$\mathrm{transport}^R_{\mathrm{loop}^{-1}}(r) = r - 1$$

$$\mathrm{transport}^{x \mapsto \star_{S^1} = x}(\mathrm{loop}^{r-1}) = \mathrm{loop}^{r-1} \cdot \mathrm{loop} = \mathrm{loop}^r$$

and we are done. This proves that $\Omega S^1 \simeq \mathbb{Z}$ (with inverse given by $\mathrm{loop}^{(-)}$).

Corollary. *S^1 is a groupoid (1-type)*

Proof.
- WTS for all $x, y : S^1$ that $x = y$ is a set.
- 'Being a set' is a propoistion – ETS when $x := y := \star_{S^1}$
- $(\star_{S^1} = \star_{S^1}) \simeq \mathbb{Z}$ which is a set. □

Corollary. *$\pi_n(S^1) = 0$ for $n > 1$*

Corollary. *$\pi_1(S^1) \cong \mathbb{Z}$.*

Proof. We have

$$\pi_1(S^1) := \|\Omega S^1\|_0 \simeq \|\mathbb{Z}\|_0 \simeq \mathbb{Z}$$

- Homomorphism? Inverse is given by $n \mapsto |\mathrm{loop}^n|_0$ and clearly

$$|\mathrm{loop}^{n+m}|_0 = |\mathrm{loop}^n \cdot \mathrm{loop}^m|_0$$ □

The long exact sequence – A powerful tool for computing new homotopy groups from already known ones

Let $f : E \to_\star B$ be a pointed map and $F := \mathrm{fib}_f(\star_B)$. Get seq. of poined maps

$$F \xrightarrow{\mathrm{fst}} E \xrightarrow{f} B$$

Furthermore, we get a map $\partial : \Omega B \to F$ defined by $\partial(p) := (\star_B, \star_f \cdot p)$.

- This gives us an infinite sequence of maps

$$\begin{tikzcd}
 & \dots \arrow[r, "\Omega(f)"] & \Omega(B) \arrow[dll, dashed] \\
\Omega(F) \arrow[r, "\Omega(\mathrm{fst})"] & \Omega(A) \arrow[r, "\Omega(f)"] & \Omega(B) \arrow[dll] \\
F \arrow[r, "\mathrm{fst}"] & A \arrow[r, "f"] & B
\end{tikzcd}$$

Applying set truncation, we get a long exact sequence (LES)

$$\begin{tikzcd}
 & \dots \arrow[r, "f_*"] & \pi_2(B) \arrow[dll, dashed] \\
\pi_1(F) \arrow[r, "\mathrm{fst}_*"] & \pi_1(A) \arrow[r, "f_*"] & \pi_1(B) \arrow[dll] \\
\pi_0(F) \arrow[r, "\mathrm{fst}"] & \pi_0(A) \arrow[r, "f_*"] & \pi_0(B)
\end{tikzcd}$$

Exactness = image of a map equals the kernel of the next.

Theorem. *There is a map* $\eta : S^3 \to S^2$ *s.t.* $\mathrm{fib}_\eta(\star_{S^2}) \simeq S^1$

Proof. Soon. □

Applying LES with $F = S^1$, $E = S^3$ and $B = S^2$, we get

$$\begin{tikzcd}
 & \pi_2(S^3) \arrow[r] & \pi_2(S^2) \arrow[dll] \\
\pi_1(S^1) \arrow[r] & \pi_1(S^3) &
\end{tikzcd}$$

i.e.

$$\begin{tikzcd}
 & 1 \arrow[r] & \pi_2(S^2) \arrow[dll] \\
\pi_1(S^1) \arrow[r] & 1 &
\end{tikzcd}$$

So $\pi_2(S^2) \cong \pi_1(S^1) \cong \mathbb{Z}$!
Also:

$$\begin{tikzcd}
\pi_3(S^1) \arrow[r] & \pi_3(S^3) \arrow[r] & \pi_3(S^2) \arrow[dll] \\
\pi_2(S^1) & &
\end{tikzcd}$$

i.e.

$$\begin{tikzcd}
1 \arrow[r] & \pi_3(S^3) \arrow[r] & \pi_3(S^2) \arrow[dll] \\
1 & &
\end{tikzcd}$$

So $\pi_3(S^2) \cong \pi_3(S^3)$. We will compute the latter group tomorrow.

- How to prove theorem?

Multiplying points on the circle

Def. An **H-space** is a pointed type X together with a product $\mu : X \times X \to X$ together with:

$$\begin{aligned}
l : (x : X) &\to \mu(\star_X, x) = x \\
r : (x : X) &\to \mu(x, \star_X) = x \\
p : l(\star_X) &= r(\star_X)
\end{aligned}$$

or, alternatively,

$$\begin{tikzcd}
X \times X \arrow[r, dotted, "\mu"] & X \\
X \vee X \arrow[u] \arrow[ur] &
\end{tikzcd}$$

Classically: $S^1 \simeq \mathrm{U}(1) = \{\, z : \mathbb{C}^2 \mid |z| = 1 \,\}$, so there is a multiplication $S^1 \times S^1 \to S^1$ which gives S^1 an H-space structure.

In HoTT we can define this multiplication by (double) pattern matching:

$$\begin{aligned} \star_{S^1} \cdot x &:= x \\ \mathrm{ap}_{(-)\cdot\star_{S^1}}(\mathrm{loop}) &:= \mathrm{loop} \\ \mathrm{apd}_{y\mapsto \mathrm{ap}_{(-)\cdot y}}(\mathrm{loop}) &:= \ldots : \mathrm{loop} =^{x\mapsto x=x}_{\mathrm{loop}} \mathrm{loop} \end{aligned}$$

exercise: fill in the gaps.

Proposition. *Ffor any $x : S^1$, multiplication $x : \cdot(-) : S^1 \to S^1$ is an equivalence.*

Proof. Statement is a prop, so ets when $x = \star_{S^1}$. But $\star_{S^1} \cdot (-) = \mathrm{id}_{S^1}$, so we are done. □

In particular, we have inverses $(-)^{-1} : S^1 \to S^1$.

The Hopf fibration

Def. We define a type family $H : S^2 \to \mathcal{U}$ by induction:

$$\begin{aligned} H(N) := H(S) &:= S^1 \\ \mathrm{ap}_H(\mathrm{merid}_z) &:= \mathsf{ua}(z \cdot _) \end{aligned}$$

The map $(x : S^2) \times H(x) \xrightarrow{\mathrm{fst}} S^2$ has fibres $H(\star_{S^2}) := S^1$. So done if we can show that $(x : S^2) \times H(x) \simeq S^3$.

S^2 is a pushout – need to examine how pushouts interact with sum types:

Lemma (Flattening lemma(ish))**.** *Given a pushout square*

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d, "\mathsf{inr}"] \\ B \arrow[r, "\mathsf{inl}"'] \arrow[ur, phantom, "\ulcorner"] & D \end{tikzcd}$$

and a dependent type $E : D \to \mathcal{U}$. The following two squares are both pushout squares:

$$\begin{tikzcd} (a : A) \times E(\mathsf{inl}(f(a))) \arrow[r] \arrow[d] & (c : C) \times E(\mathsf{inr}(x)) \arrow[d] \\ (b : B) \times E(\mathsf{inl}(b)) \arrow[r] \arrow[ur, phantom, "\ulcorner"] & (x : D) \times E(x) \end{tikzcd}$$

$$\begin{tikzcd} (a : A) \times E(\mathsf{inr}(g(a))) \arrow[r] \arrow[d] & (c : C) \times E(\mathsf{inr}(x)) \arrow[d] \\ (b : B) \times E(\mathsf{inl}(b)) \arrow[r] \arrow[ur, phantom, "\ulcorner"] & (x : D) \times E(x) \end{tikzcd}$$

Since the suspension ΣA is the pushout of the span $1 \leftarrow A \rightarrow 1$, this applies to the family H, and we conclude that the total space $(z : \Sigma A) \times H(z)$ is the pushout of the span at the top of the following map of spans:

$$\begin{tikzcd}
A \arrow[d, "\mathrm{id}"'] & A \times A \arrow[l, "\mathrm{fst}"'] \arrow[r, "(a,b)\mapsto a\cdot b"] \arrow[d, "(a,b)\mapsto(a,a\cdot b)"] & A \arrow[d, "\mathrm{id}"] \\
A & A \times A \arrow[l, "\mathrm{fst}"'] \arrow[r, "\mathrm{snd}"] & A
\end{tikzcd}$$

Since the vertical maps are equivalences, the pushouts are equivalent, so the total space is equivalent to $A * A$.

Proposition. *For the Hopf fibration $H : S^2 \rightarrow \mathcal{U}$, the total space can be identified with S^3.*

Proof. Use $S^1 * S^1 = S^3$ from previous lecture. Since S^2 is the pushout of $1 \leftarrow S^1 \rightarrow 1$, we can use flattning to deduce that the following is a pushout square:

$$\begin{tikzcd}
S^1 \times S^1 \arrow[dr, equal] & & S^1 \arrow[d, equal] \\
& S^1 \times H(N) \arrow[r] \arrow[d] & 1 \times H(S) \arrow[d] \\
S^1 \arrow[r, equal] & 1 \times H(N) \arrow[r] & (x : S^2) \times H(x) \arrow[ul, phantom, "\ulcorner"]
\end{tikzcd}$$

But $S^1 * S^1$ is also the pushout of this span. $\square$

Lecture 4: The Freudenthal Suspension Theorem and Degrees

In this chapter we'll compute some nontrivial homotopy groups of spheres, namely $\pi_n(S^n) = \mathbb{Z}$ for $n \geq 2$, from which we directly get $\pi_3(S^2) = \mathbb{Z}$.

Our main tool is the Freudenthal Suspension Theorem:

Theorem (Freudenthal)**.** *If X is a pointed n-connected type with $n \geq 0$, then the unit map $\eta : X \to_\star \Omega\Sigma X$ is $2n$-connected.*

There is a hands-on proof in the HoTT book (Univalent Foundations Program 2013), based on the wedge connectivity lemma. This proof is a bit mysterious (to me), so instead we'll derive it from another celebrated result, namely the Blakers–Massey theorem:

Theorem (Blakers–Massey)**.** *In any pushout square of maps f and g,*

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] \arrow[dr, phantom, "\ulcorner"] & C \arrow[d, "\mathrm{inr}"] \\ B \arrow[r, "\mathrm{inl}"'] & Q, \end{tikzcd}$$

where f is n-connected and g is m-connected, if we form the pullback P of the cospan $B \to Q \leftarrow C$, then we get an induced map (the gap map) $d : A \to P$, and this is $(n+m)$-connected.

Now **??** follows directly from **??** via the pushout square

$$\begin{tikzcd} X \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & 1 \arrow[d] \\ 1 \arrow[r] & \Sigma X, \end{tikzcd}$$

whose gap map is the meridian map, $\mathrm{merid} : X \to N =_{\Sigma X} S$, which is identified with $\eta : X \to \Omega\Sigma X$ via composition with $\mathrm{merid}_{\mathrm{pt}_X}$.

We defer the proof of **??** a bit, preferring instead first to harvest the fruits of **??** .

Corollary. *Suspension induces an isomorphism $\pi_k(S^n) \to \pi_{k+1}(S^{n+1})$ for $k \leq 2n-2$.*

Def. The **stable homotopy groups of spheres** are the groups $\pi_k(\mathbb{S}) := \pi_{k+n}(S^n)$ for $n \geq k+2$.

Note that the Freudenthal suspension theorem by itself doesn't quite suffice to show that $\pi_1(S^1) \to \pi_2(S^2)$ is an equivalence. But we already know $\pi_2(S^2) = \mathbb{Z}$, so we get:

Corollary. *We have $\pi_n(S^n) = \mathbb{Z}$ for $n \geq 1$.*

Whitehead's theorem and principle

There's another way to get **??** , which is illuminating in itself, and involves a bit more careful analysis of n-connected maps.

This is closely related to Whitehead's theorem and principle. In the classical model of infinity groupoids, this is true for all types, but there are many models where that version fails:

Theorem (Whitehead)**.** *Suppose X and Y are n-types and $f : X \to Y$ induces a bijection on components, $\|X\|_0 \to \|Y\|_0$, and an isomorphism $\pi_k(X, x) \to \pi_k(Y, f(x))$ for all $x : X$ and all $1 \leq k \leq n$. Then f is an equivalence.*

Proof. By induction on n, the case $n = -2$ being trivial.

In the step case, it suffices to show $\Omega(f) : \Omega(X, x) \to \Omega(Y, f(x))$ is an equivalence for all $x : X$. This follows by induction hypothesis and the generalized fact that $\pi_k(\mathrm{ap}_f) : \pi_k(x =_X x', q) \to \pi_k(f(x) =_Y f(x'), \mathrm{ap}_f(q))$ is an isomorphism for $1 \leq k < n$ and all $x' : X$ and $q : x =_X x'$, which follows by path induction on q and then the hypothesis. □

Corollary. *An n-type X is contractible if and only if X is connected and $\pi_k(X, x)$ vanishes for all $1 \leq k \leq n$ and $x : X$.*

Corollary. *For $n \geq 0$, a map $f : X \to Y$ is n-connected if and only if we have:*

- *$\|f\|_0 : \|X\|_0 \to \|Y\|_0$ is an isomorphism;*
- *$\pi_k(f) : \pi_k(X, x) \to \pi_k(X, f(x))$ is an isomorphism for $1 \leq k \leq n$ and $x : X$,*
- *$\pi_{n+1}(f)$ is surjective for all $x : X$.*

Proof. The "only if" part comes from the LES, and the "if" part from the LES and **??** applied to the fibers of f. □

NB It's possible to prove **??** without using the LES, and this then gives a more direct proof that suspension is an equivalence $\pi_1(S^1) \to \pi_2(S^2)$.

Join and wedge connectivity

Both Freudenthal and Blakers–Massey were originally proved in HoTT by clever arguments using the wedge connectivity lemma (Lumsdaine and Licata 2012; Hou (Favonia), Finster, Licata, and Lumsdaine 2016). The wedge connectivity lemma itself comes from the dual Blakers–Massey theorem, which is quite a bit easier than Blakers–Massey itself:

Theorem (Dual Blakers–Massey)**.** *In any pullback square of maps f and g,*

$$\begin{tikzcd} X \times_Z Y \arrow[r, "\mathrm{snd}"] \arrow[d, "\mathrm{fst}"'] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & Z, \end{tikzcd}$$

where f is n-connected and g is m-connected, if we form the pushout Q of the span $X \leftarrow X \times_Z Y \rightarrow Y$, then we get an induced map (the cogap) $c : Q \rightarrow Z$, and this is $(n + m + 2)$-connected.

This in turns follows from the join connectivity lemma:

Lemma (Join connectivity)**.** *If X is n-connected and Y is m-connected, then $X * Y$ is $(n + m + 2)$-connected.*

Proof sketch. Suppose Z is an $(n + m + 2)$-type. Then $Z \rightarrow (X * Y \rightarrow Z)$ is an equivalence if and only if

$$Z \rightarrow (X \rightarrow Z) \times_{(X \times Y \rightarrow Z)} (Y \rightarrow Z)$$

is, which in turn means that we have unique diagonal lifts in the square

$$\begin{tikzcd} X \arrow[r] \arrow[d] & Z \arrow[d] \\ 1 \arrow[r] \arrow[ur, dashed] & (Y \rightarrow Z). \end{tikzcd}$$

So it suffices to check that $Z \rightarrow (Y \rightarrow Z)$ is n-truncated. So we look at the diagonal fillers in a square

$$\begin{tikzcd} S^{n+1} \arrow[r] \arrow[d] & Z \arrow[d] \\ 1 \arrow[r] \arrow[ur, dashed] & (Y \rightarrow Z), \end{tikzcd}$$

which are unique if $Z \rightarrow (S^{n+1} * Y \rightarrow Z)$ is an equivalence. But $S^{n+1} * Y \simeq \Sigma^{n+2} Y$ is $(n + m + 2)$-connected, so this checks out. □

See Anel, Biedermann, Finster, and Joyal (2020) for the most streamlined approach to these matters.

The zigzag construction

The best explanation we have (in my opinion) of the Blakers–Massey theorem comes from the work of Wärn (2024) on the zigzag construction. This gives us precise information on the path spaces of pushouts, as follows.

Consider a span, straightened out as a family $R : A \to B \to \mathcal{U}$, giving a pushout square:

$$\begin{tikzcd} (x : A) \times (y : B) \times R(x, y) \arrow[r] \arrow[d] & B \arrow[d, "\mathrm{inr}"] \\ A \arrow[r, "\mathrm{inl}"'] \arrow[ur, phantom, "\ulcorner", very near start] & Q \end{tikzcd}$$

We want to understand the identity types $\mathrm{inl}(a_0) = \mathrm{inl}(x)$ and $\mathrm{inl}(a_0) = \mathrm{inr}(y)$ for $x : A$ and $y : B$.

The initial observation, due to Kraus and Raumer (2021), is that these are freely generated by $\mathrm{refl} : \mathrm{inl}(a_0) = \mathrm{inl}(a_0)$ and equivalences $r \cdot _ : (\mathrm{inl}(a_0) = \mathrm{inl}(x)) \simeq (\mathrm{inl}(a_0) = \mathrm{inr}(y)$ for all $r : R(x, y)$.

David Wärn then observed that we can unravel this as sequential colimits over zigzags of length at most t. He defined types $a_0 \rightsquigarrow_t x$ (t even) and $a_0 \rightsquigarrow_t y$ (t odd), $t \geq -1$. We take $a_0 \rightsquigarrow_0 x := (a_0 =_A x)$ and $a_0 \rightsquigarrow_{-1} y := 0$. In the step cases, we have pushouts

$$\begin{tikzcd} (y : B) \times R(x, y) \times (a_0 \rightsquigarrow_t x) \arrow[r, "r \cdot _"] \arrow[d] & (y : B) \times R(x, y) \times (a_0 \rightsquigarrow_{t+1} y) \arrow[d, "r^{-1} \cdot _"] \\ (a_0 \rightsquigarrow_t x) \arrow[r] \arrow[ur, phantom, "\ulcorner", very near start] & (a_0 \rightsquigarrow_{t+2} x) \end{tikzcd}$$

and similarly for $(a_0 \rightsquigarrow_{t+2} y)$.

Theorem (Wärn)**.** *We have equivalences for all $x : A$ and $y : B$:*

$$(a_0 \rightsquigarrow_\infty x) \simeq (\mathrm{inl}(a_0) =_Q \mathrm{inl}(x))$$
$$(a_0 \rightsquigarrow_\infty y) \simeq (\mathrm{inl}(a_0) =_Q \mathrm{inr}(y)),$$

where $(a_0 \rightsquigarrow_\infty x) := \varinjlim_t (a_0 \rightsquigarrow_{2t} x)$ and $(a_0 \rightsquigarrow_\infty y) := \varinjlim_t (a_0 \rightsquigarrow_{2t-1} y)$.

Corollary (van Kampen)**.** *The groupoid $\|Q\|_1$ can be described with identity types as set quotients of sequences. E.g., $\|\mathrm{inl}(a_0) = \mathrm{inl}(x)\|_0$ is the set quotient of sequences*

$$(a_0, r_0, b_0, r_1, a_1, \ldots, b_n, r_{n+1}, x),$$

modulo erasing steps $(r, _, r)$.

Corollary. *Suppose A and B are 1-types and both maps in the span $A \leftarrow R \to B$ are 0-truncated. Then Q is a 1-type and the gap map is an embedding.*

This solved a long-standing open problem, showing that the free ∞-group of a set is in fact a 1-group.

We also get **??** from the following more precise theorem:

Theorem. *Suppose in the span $A \leftarrow R \rightarrow B$ the legs are n- and m-connected, respectively. Then for any $r : R(x, y)$, the map $r \cdot _ : (a_0 \rightsquigarrow_{2k} x) \rightarrow (a_0 \rightsquigarrow_{2k+1} y)$ is $(k(n+m+4)+m)$-connected, and the map $r^{-1} \cdot _ : (a_0 \rightsquigarrow_{2k-1} y) \rightarrow (a_0 \rightsquigarrow_{2k} x)$ is $(k(n+m+4)-2)$-connected.*

Chapter 1

Outlook

Things we didn't talk about, in no particular order:

- Projective spaces (Buchholtz and Rijke 2017).
- Eilenberg–Mac Lane spaces (Licata and Finster 2014; Buchholtz, J. Daniel Christensen, Flaten, and Rijke 2023; Wärn 2023).
- $\pi_4(S^3)$ via Whitehead products and the Gysin sequence in Brunerie's thesis (Brunerie 2016). Later different computation by Ljungström and Mörtberg (2023).
- The EHP sequence (Cagne, Buchholtz, Kraus, and Bezem 2024).
- Spectra and spectral sequences (Doorn 2018).
- Cohomology (Buchholtz and Hou (Favonia) 2020; Lamiaux, Ljungström, and Mörtberg 2023).
- Homology (Graham 2018),
- The Hurewicz Theorem (J Daniel Christensen and Scoccola 2023).
- Steenrod operations (Brunerie 2017).
- Syllepsis (Sojakova and Kavvos 2022).
- Nilpotent types (Scoccola 2020).
- Higher group theory (Buchholtz, Doorn, and Rijke 2018; Buchholtz and Rijke 2023; Bezem, Buchholtz, Cagne, Dundas, and D. R. Grayson 2024; Swan 2022).
- Modalities and localization (Rijke, Shulman, and Spitters 2020; J. Daniel Christensen, Opie, Rijke, and Scoccola 2020; J. Daniel Christensen and Rijke 2022; Myers 2022; Myers and Riley 2023).
- …

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